SILTHAUL · PROJECT PLAN

SiltHaul project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First project plan, written for the kit 1.9.1 upgrade

SiltHaul project plan

SiltHaul is at stage P3 (design outputs, TRL 3) with three modelled concept options; the next gate is Amish's choice between option B and option A (SLH-OPT-001, SLH-DDR-005), and TRL 4 work stays on hold while the portfolio cap is TRL 3. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem, users and stakeholders	Project lead	SLH-PRB-001 with user and stakeholder tables	Done (P3)
STEM check	Design lead	SLH-STM-001, `stem_check.py`; independent review recorded	Done; 12 plausible rows open to TRL 4
Concept options	Design lead	SLH-OPT-001 and the options picture	Done; choice awaiting Amish
Requirements and traceability	Design lead	SLH-URS-001, SLH-PRS-001, SLH-RTM-001	Done; to review with a co-design partner
Calculations	Design lead	SLH-CAL-001, `sizing.py`, `options_calc.py`	Done for A; first-order for B and C
CAD at D3, every option	Design lead	`cad/src/model.py`, STEP per part and per assembly, `cad/fidelity.json`	Done
Drawings, BOM, build plan	Design lead	SLH-DWG-001 and 002, BOM, SLH-BLD-001	Done for option A; B follows only if Amish chooses it
Regulatory, feasibility, risk, market	Project lead	SLH-REG-001, FEA-001, RSK-001, MKT-001	Done (first pass)
Co-design partner	Project lead	Agreement with a disaster authority, NGO or volunteer group	Not started; Amish decides
Prototype build, tests and trials (P4)	Fabricator, test lead	TST plans and reports	On hold: TRL 4
Field validation (P5)	Co-design partner	Validation and human factors reports	On hold
Launch (P6)	Project lead	Instructions for use, training	On hold

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Local fabricator	First build of the chosen option	To assess	Parts 1,055.60 for option A (BOM); labour to assess
Test lead and a test bed of wet silt	Timed trial against a bucket crew (PR-001), release and anchor tests	To assess	To assess
Proof-load rig (CalRig, shared block)	Anchor, rope and sling proof loads (PR-006, PR-007)	To assess	To assess
Regulatory consultant	Machinery rules of the country of use (SLH-REG-001 G9)	To assess	To assess

No cost is invented here: the only priced item is the BOM, at the indicative prices in `bom/bom.csv`.

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	Problem, STEM check, options, first REG, FEA, PLN	Done
P1 Definition	URS, PRS, RSK, MKT, REG complete	Done
P2 Design inputs	PRS with targets, CAL, RTM, concept selection DDR, STEM check closed	Done except the selection: awaiting Amish; STEM check open (12 plausible rows)
P3 Design outputs	Model, STEP, drawings, priced BOM, build plan, decisions register, design FMEA, traceability complete	Reached 2026-11-03
P4 Verification	Test plans, lab reports, verification report	On hold (TRL 4)
P5 Validation	Field trials with a partner	On hold
P6 Commercialization	Launch plan, training	On hold; needs a named partner

Next steps and decision requested

1. Amish chooses between option B and option A, or asks for a hybrid (SLH-DDR-005, SLH-OPT-001). Nothing is chosen here.
2. If Amish chooses B, the BOM, drawings and build plan are redone for B; if A, nothing changes.
3. Amish decides whether to approach a co-design partner and whether to move to TRL 4 when the portfolio cap is lifted. The review note recommends it only after the choice.

References

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